

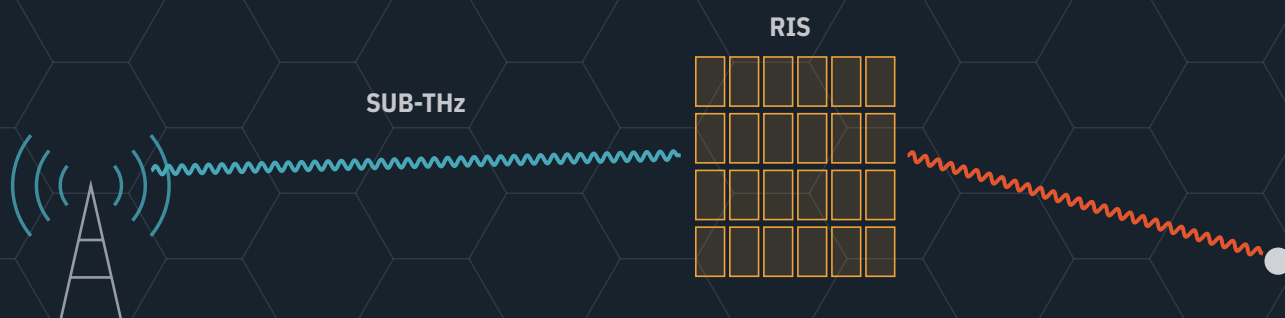
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 26

Future 6G Technologies for Advanced Communication Networks

فناوری‌های آینده 6G برای شبکه‌های ارتباطی پیشرفته



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

In This Module

⚠ IMPORTANT DISCLAIMER	2
Overview: The Road to 6G	3
1 AI-Native Networks / AI-RAN	4
2 Extremely Large MIMO (XL-MIMO)	6
3 Cell-Free Massive MIMO	8
4 Integrated Sensing and Communication (ISAC / JCS)	10
5 Reconfigurable Intelligent Surfaces (RIS)	12
6 Sub-THz / THz Communications	15
7 Non-Terrestrial Networks (NTN)	17
8 Semantic Communications	19
9 Digital Twins for Networks	22
10 Advanced Positioning (cm-Level)	24
Technology Maturity Matrix	27
Projected Timeline: 3GPP Releases and 6G Technologies	28
Summary: 5G vs 6G Paradigm Shifts	29
Key Takeaways	29
References and Further Reading	30

⚠ IMPORTANT DISCLAIMER

This module discusses technologies at various stages of maturity. Many concepts presented here are speculative, under early research, or in preliminary standardization study phases. The 6G standard does not yet exist — ITU IMT-2030 framework is under development, and 3GPP has not begun formal 6G specification work (expected ~2025-2028). Information is based on published research, 3GPP study items, and industry roadmaps as of 2024-2025.

■ Maturity Classification Used in This Module

Label	Meaning
STANDARDIZED	Specified in 3GPP Technical Specifications, deployable
STUDY ITEM	Formally studied in 3GPP (TR documents), may become work items
CANDIDATE	Strong industry consensus, likely to enter standardization
RESEARCH	Active academic/industry research, no formal standardization yet
SPECULATIVE	Early-stage concepts, feasibility not fully demonstrated

Overview: The Road to 6G

■ IMT-2030 Vision (ITU-R)

The ITU-R Working Party 5D has outlined the IMT-2030 framework with these usage scenarios:

- **Immersive Communication** (evolution of eMBB)
- **Hyper Reliable and Low-Latency Communication** (evolution of URLLC)
- **Massive Communication** (evolution of mMTC)
- **Ubiquitous Connectivity** (new)
- **Integrated AI and Communication** (new)
- **Integrated Sensing and Communication** (new)

5G Capabilities vs 6G Targets (IMT-2030)

KPI	5G (IMT-2020)	6G Target (IMT-2030)	Improvement
Peak data rate	20 Gbps	200+ Gbps	10×
User experienced rate	100 Mbps	1-10 Gbps	10-100×
Latency (user plane)	1 ms	0.1 ms	10×
Reliability	99.999%	99.99999%	100×
Connection density	10^6 devices/km ²	10^7 - 10^8 devices/km ²	10-100×
Mobility	500 km/h	1000+ km/h	2×
Spectrum efficiency	1× (baseline)	3-5×	3-5×
Energy efficiency	1× (baseline)	10-100×	10-100×
Positioning accuracy	~1 m (R16)	1-10 cm	10-100×
AI integration	External/OTT	Native/intrinsic	Paradigm shift
Sensing capability	Not supported	Integrated	New capability

1 AI-Native Networks / AI-RAN

Status: ● STUDY ITEM (Release 18)

What

AI-Native Networks integrate artificial intelligence as a fundamental design principle rather than an add-on optimization layer. AI/ML algorithms are embedded directly into the radio access network (AI-RAN) at every protocol layer.

Why / Problem Addressed

- Traditional model-based algorithms (MMSE, LMMSE) reach performance limits in complex channels
- Manual network optimization cannot scale to ultra-dense heterogeneous deployments
- 5G NR parameter space is enormous (beamforming, scheduling, power control)
- Need for self-optimizing, self-healing networks

AI Integration Across Protocol Layers

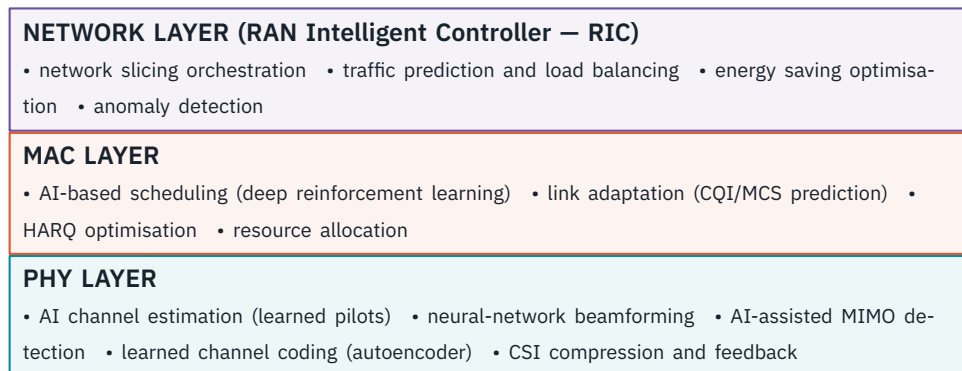


Figure 26.1 AI across the protocol layers

3GPP Progress

- **TR 38.843:** Study on AI/ML for NR air interface (Release 18)
- **Use cases studied:**
 - CSI feedback enhancement (AI-based compression)
 - Beam management (AI-based beam prediction)
 - Positioning enhancement (AI/ML-assisted)
- **TR 38.844:** Study on AI/ML for NG-RAN (network level)
- **Key finding:** Performance gains demonstrated but deployment complexity is high

Benefits

- 10-30% throughput improvement in complex scenarios
- Reduced CSI feedback overhead (up to 50% compression)
- Faster beam acquisition (reduced beam sweeping)
- Self-optimizing network behavior

Challenges

- **Training data:** Requires massive, representative datasets
- **Generalization:** Models may not transfer across environments
- **Computational cost:** Inference latency vs. real-time requirements
- **Standardization:** How to specify AI models in a standard?
- **Interoperability:** Models from different vendors must coexist
- **Explainability:** Black-box decisions in safety-critical scenarios

Relation to 5G

- 5G O-RAN architecture provides RIC framework (xApps/rApps)
- R18 studies lay groundwork; actual specifications expected R19-R20
- Evolution from “AI-assisted” (5G) to “AI-native” (6G)

Open Questions

- How to handle model updates over-the-air?
- What is the right split between model-based and data-driven?
- How to ensure robustness against adversarial attacks?
- Standardize the model or just the interface?

2 Extremely Large MIMO (XL-MIMO)

Status: ● RESEARCH

What

XL-MIMO scales antenna arrays to thousands or tens of thousands of elements, creating electrically very large apertures. Unlike conventional massive MIMO (64-256 elements), XL-MIMO operates partly in the **near-field** regime, fundamentally changing propagation characteristics.

Why / Problem Addressed

- Massive MIMO gains saturate with conventional array sizes
- Need for extreme spatial multiplexing (100+ simultaneous streams)
- Higher frequencies require larger arrays for equivalent beamforming gain
- Support for extremely high data rates in 6G

Key Physics: Near-Field vs Far-Field

Far-field boundary (Fraunhofer distance):

$$d_F = \frac{2D^2}{\lambda}$$

Example: a $D = 2$ m array at 30 GHz ($\lambda = 10$ mm) gives $d_F = \frac{2 \times 4}{0.01} = 800$ m, so **most users are in the NEAR FIELD of large arrays!**

Near-field characteristics:

- Spherical wavefronts (not planar)
- Beam focusing (not just steering) — energy concentrated at a point in space
- Distance-dependent beamforming

- Spatial non-stationarity: not all array elements “see” the same channel

■ Spatial Non-Stationarity

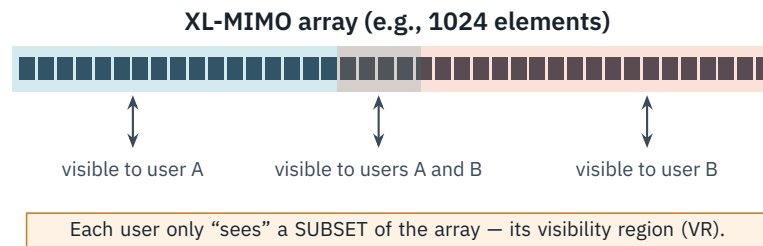


Figure 26.2 Spatial non-stationarity: each user sees only part of the array

■ Benefits

- Beam focusing enables spatial division even for users at same angle
- Near-field MIMO provides additional degrees of freedom
- Extreme spectral efficiency through massive spatial multiplexing
- Inherent interference suppression

■ Challenges

- Channel estimation complexity: $O(N^2)$ pilot overhead without structure exploitation
- Hardware: cost, power, interconnect for thousands of RF chains
- Signal processing: centralized processing infeasible → distributed architectures
- Channel modeling: 3GPP models assume far-field; new models needed
- Calibration: mutual coupling, array imperfections at scale

■ Relation to 5G

- Extension of 5G massive MIMO (R15-R17: up to 256 elements)
- 5G codebook-based and Type-II CSI approaches won't scale
- Requires new channel models beyond 3GPP TR 38.901

■ Open Questions

- Optimal array geometry (planar, cylindrical, conformal)?
- How to exploit near-field properties in standard protocols?
- Hybrid analog/digital architectures at this scale?
- Energy-efficient implementations?

3 Cell-Free Massive MIMO

■ **Status:** ● RESEARCH (5G-Advanced precursors exist)

■ What

Cell-Free Massive MIMO eliminates the concept of cells. A large number of distributed Access Points (APs), connected via fronthaul to Central Processing Units (CPUs), coherently serve all User Equipments (UEs) in the area without cell boundaries.

■ Why / Problem Addressed

- Cell-edge problem: users at cell boundaries suffer from interference and low SINR
- Uneven QoS distribution in cellular networks (10th-percentile vs median)
- Inter-cell interference is a fundamental limitation of cellular architecture
- Need for uniform, ubiquitous coverage

■ Architecture

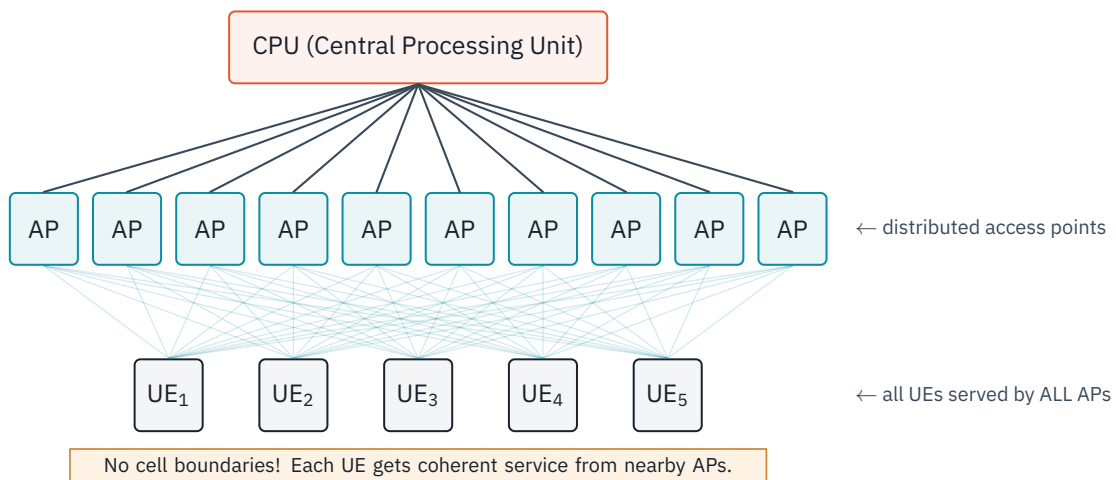


Figure 26.3 Cell-free massive MIMO: all APs serve all UEs

■ Key Principles

1. **All APs serve all UEs** (in principle; practical systems use AP selection)
2. **Coherent joint transmission/reception** across distributed APs
3. **No handover** — UE is always served by the “best” cluster of APs
4. **User-centric clustering** — each UE has its own virtual cell

Benefits

- 95th-percentile throughput improvement: $5\text{-}10\times$ over cellular
- Elimination of cell-edge effect
- Macro-diversity: robust against blockage and shadowing
- Natural interference management through coordination
- Uniform coverage quality

Challenges

- **Fronthaul requirements:** Massive data exchange between APs and CPU
- **Synchronization:** All APs must be phase-synchronized for coherent transmission
- **Scalability:** Computational complexity grows with number of APs $\times$ UEs
- **Channel estimation:** Pilot contamination across large areas
- **Practical deployment:** Cost of dense AP infrastructure

Scalable Implementations

Level	Processing	Fronthaul	Performance
Level 1	Fully centralized MMSE	Very high	Maximum
Level 2	Local MMSE + central combining	High	Near-optimal
Level 3	Local MF + central LSFD	Moderate	Good
Level 4	Local MF, no cooperation	Low	Baseline

Relation to 5G

- 5G CoMP (Coordinated MultiPoint) is a limited precursor
- 5G-Advanced (R18): Multi-TRP enhancements move toward cell-free concepts
- C-RAN architecture provides infrastructure basis
- O-RAN fronthaul interfaces (F1, eCPRI) enable distributed processing

Open Questions

- Optimal AP density and placement?
- How to handle mobility efficiently?
- Power control across distributed APs?
- Economic viability vs. traditional deployment?

4 Integrated Sensing and Communication (ISAC / JCS)

■ **Status:** ● **STUDY ITEM (Release 19)**

■ What

ISAC (also called Joint Communication and Sensing, JCS) uses a shared waveform, shared hardware, and shared spectrum to simultaneously perform wireless communication and radar-like sensing (detection, ranging, velocity estimation, imaging).

■ Why / Problem Addressed

- Separate radar and communication systems waste spectrum and hardware
- Autonomous vehicles, drones, industrial robots need both connectivity and environmental awareness
- Spectrum scarcity drives dual-use of allocated bands
- 5G NR waveform (OFDM) is already suitable for sensing with modifications

■ Sensing Capabilities

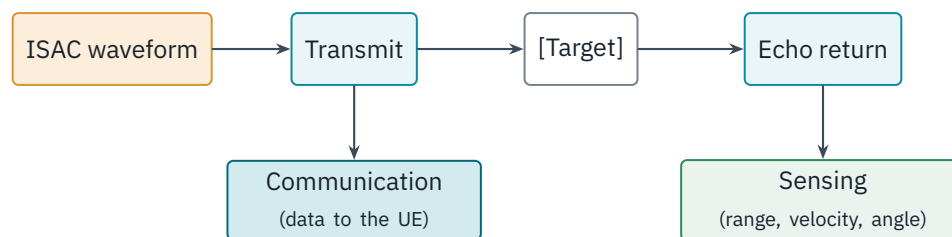


Figure 26.4 Integrated sensing and communication with one waveform

Sensing parameters extracted:

- **Range:** $R = c\tau/2$ (from the delay τ)
- **Velocity:** $v = \lambda f_d/2$ (from the Doppler shift f_d)
- **Angle:** θ from array processing (AoA/AoD)

OFDM Radar Processing

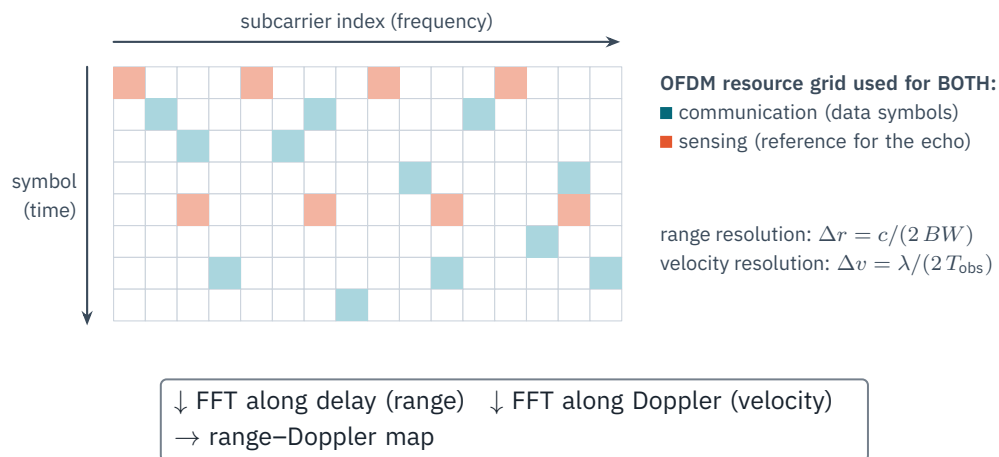


Figure 26.5 OFDM radar processing on the resource grid

3GPP Progress

- **R19 Study Item:** “Study on Integrated Sensing and Communication” (approved 2024)
- Key study areas:
 - Channel model for sensing (bistatic, monostatic)
 - Waveform design and resource allocation
 - Sensing-assisted communication (e.g., beam prediction from sensing)
 - Privacy and regulatory aspects of network-based sensing
- Use cases: gesture recognition, object tracking, intrusion detection, weather monitoring

Benefits

- Spectrum efficiency: dual use of same resources
- Hardware reuse: shared antenna arrays and RF chains
- Synergy: sensing assists communication (predictive beamforming) and vice versa
- New services: network as a sensor (NaaS — Network as a Sensor)

Challenges

- **Waveform tradeoff:** Optimal for communication $\neq$ optimal for sensing
- **Self-interference:** Monostatic sensing requires full-duplex capability
- **Clutter:** Distinguish targets from environment
- **Privacy:** Network can sense people without their devices
- **Regulation:** Sensing in licensed communication bands needs new frameworks
- **Performance metrics:** How to jointly optimize communication rate and sensing accuracy?

■ Relation to 5G

- 5G NR positioning (R16/R17) is a limited precursor
- 5G OFDM waveform is sensing-capable with proper processing
- R19 study prepares for R20 work items
- 5G-Advanced adds framework; 6G makes it native

■ Open Questions

- Monostatic vs bistatic vs networked sensing — which architecture?
- How to allocate resources between sensing and communication optimally?
- What sensing performance is achievable with communication waveforms?
- Regulatory framework for passive sensing of people?

5 Reconfigurable Intelligent Surfaces (RIS)

■ Status: ● STUDY ITEM (Release 18-19)

■ What

Reconfigurable Intelligent Surfaces (RIS) are planar structures composed of many sub-wavelength elements that can be electronically controlled to reflect (or refract) incoming electromagnetic waves in desired directions. They reshape the wireless propagation environment without generating new signals.

■ Why / Problem Addressed

- mmWave/sub-THz signals suffer from severe blockage
- Traditional relays require full RF chains (expensive, power-hungry)
- NLoS coverage gaps in dense urban environments
- Need for energy-efficient coverage extension

Operating Principle

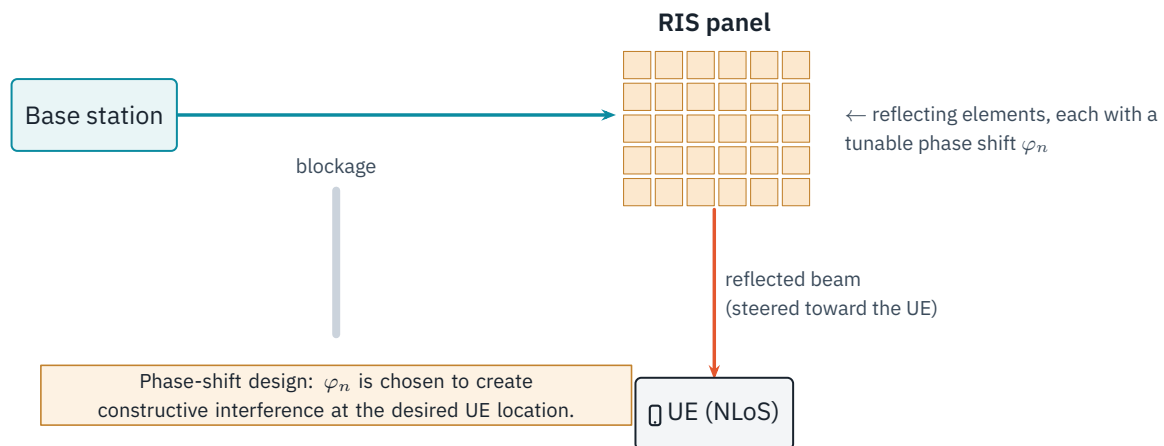


Figure 26.6 A reconfigurable intelligent surface steers the beam to a blocked UE

Types of RIS

Type	Characteristics	Gain	Complexity
Passive RIS	Only phase shift, no amplification	Moderate	Low
Active RIS	Phase shift + amplification	High	Medium
STAR-RIS	Simultaneous transmit and reflect	High	High
Hybrid RIS	Some active, some passive elements	Configurable	Medium

Path Loss Model

For a passive RIS with N elements:

$$P_{\text{Rx}} \propto N^2 \times L_{\text{BS} \rightarrow \text{RIS}} \times L_{\text{RIS} \rightarrow \text{UE}}$$

- “Product-distance” path loss (multiplicative, not additive in dB)
- → the RIS must be placed near the BS or near the UE for best performance
- → the SNR gain scales as N^2 (coherent combining)

3GPP Progress

- **TR 38.857:** Study on NR network-controlled repeaters (R18) — includes smart repeater concepts
- **R19:** Further study on RIS-assisted NR
- Study areas: channel modeling with RIS, deployment scenarios, control signaling

■ Benefits

- Nearly passive operation (only control circuit needs power)
- Low cost per element (printed circuit technology)
- Flexible deployment (walls, ceilings, lamp posts)
- Coverage extension without backhaul
- Green communication: minimal energy consumption

■ Challenges

- **Channel estimation:** Must estimate BS→RIS and RIS→UE channels separately
- **Control overhead:** Configuring hundreds/thousands of elements in real-time
- **Narrowband response:** Phase shifts are typically frequency-flat → wideband limitation
- **Deployment:** Optimal placement, orientation, density
- **Multiplicative path loss:** Double path loss can limit practical gains
- **Hardware impairments:** Discrete phase shifts, element coupling

■ Relation to 5G

- 5G network-controlled repeaters (R18) are a precursor
- 5G mmWave deployments highlight the blockage problem RIS addresses
- R18-R19 studies inform potential R20 specifications
- May complement (not replace) small cells and relays

■ Open Questions

- How many elements needed for meaningful gain in practice?
- Who controls the RIS — operator, third party, autonomous?
- How to handle mobility (real-time reconfiguration speed)?
- Passive vs active: which wins in cost-benefit analysis?
- Can RIS work effectively for broadband (wideband) signals?

6 Sub-THz / THz Communications

■ Status: ● RESEARCH

■ What

Sub-THz (100-300 GHz) and THz (300 GHz - 3 THz) communications exploit vast unused spectrum for extreme data rates. Channels of 10+ GHz bandwidth become available, potentially enabling 100+ Gbps per link.

■ Why / Problem Addressed

- mmWave bandwidth (400 MHz - 2 GHz) insufficient for 6G peak rate targets (200+ Gbps)
- Spectrum below 100 GHz is increasingly congested
- Short-range applications (data kiosks, intra-device communication) need extreme throughput
- Backhaul/fronthaul for dense networks needs wireless alternatives to fiber

■ Frequency Bands of Interest

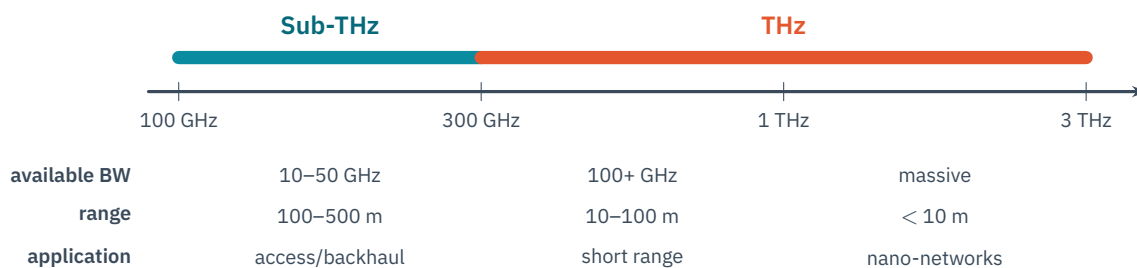


Figure 26.7 Sub-THz and THz bands: bandwidth, range and applications

Key atmospheric absorption windows:

- 130–175 GHz (moderate absorption)
- 200–320 GHz (multiple windows between absorption lines)
- water-vapour absorption lines at 183 and 325 GHz

■ Link Budget Challenge

Free-space path loss at 300 GHz versus 30 GHz (same distance, 100 m):

$$\text{FSPL} = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right)$$

- at 30 GHz: FSPL = 91.5 dB
- at 300 GHz: FSPL = 111.5 dB (+20 dB!)
- molecular absorption adds a further 1–10 dB/100 m (frequency-dependent)

Compensation requires very high-gain antennas (pencil beams), large arrays (small $\lambda \rightarrow$ many elements in a small area) and short distances.

■ Key Technologies Enabling THz

1. **III-V semiconductors** (InP, GaN): Power amplifiers up to 300 GHz
2. **Silicon-based (CMOS/SiGe)**: Lower cost, lower power, up to ~200 GHz
3. **Photonic approaches**: Optical-to-THz conversion
4. **Graphene-based devices**: Theoretical potential for THz electronics
5. **Massive arrays**: 1000+ elements possible due to small wavelength

■ Benefits

- Enormous available bandwidth (10-100+ GHz channels)
- Very small wavelength enables massive arrays in compact form factor
- Inherent security (highly directional, short range)
- High-resolution sensing capability (sub-mm resolution)

■ Challenges

- **Severe path loss**: Requires LoS and very short range
- **Atmospheric absorption**: Frequency-selective fading from molecular resonances
- **Hardware**: Limited output power, high noise figure at THz frequencies
- **Phase noise**: Oscillator stability at hundreds of GHz
- **Blockage**: Even a hand can block a THz link completely
- **Cost**: Exotic semiconductor processes

■ Relation to 5G

- 5G FR2 (24-52 GHz) demonstrated viability of high-frequency mobile access
- Lessons from 5G mmWave beam management applicable to sub-THz
- WRC-23 identified bands above 100 GHz for study (not yet allocated for mobile)
- 6G may initially target 100-300 GHz (sub-THz) before moving higher

■ Open Questions

- Is mobile access viable above 100 GHz, or only fixed/short-range?
- What is the maximum practical range for useful communication?
- How to handle beam tracking at THz beam widths ($<1^\circ$)?
- Can silicon-based solutions achieve sufficient power at these frequencies?

7 Non-Terrestrial Networks (NTN)

■ **Status:** ● **STANDARDIZED (basic R17)** / ● **STUDY (advanced features)**

■ What

Non-Terrestrial Networks integrate satellite and aerial platforms into the 3GPP network architecture. NR and IoT-NTN enable direct communication between standard UEs (including smartphones) and LEO/MEO/GEO satellites or High-Altitude Platform Systems (HAPS).

■ Why / Problem Addressed

- ~90% of Earth's surface lacks terrestrial mobile coverage
- Maritime, aviation, and remote areas have no connectivity
- Disaster resilience requires non-terrestrial backup
- Global IoT connectivity (agriculture, logistics, environmental monitoring)

■ NTN Platform Types

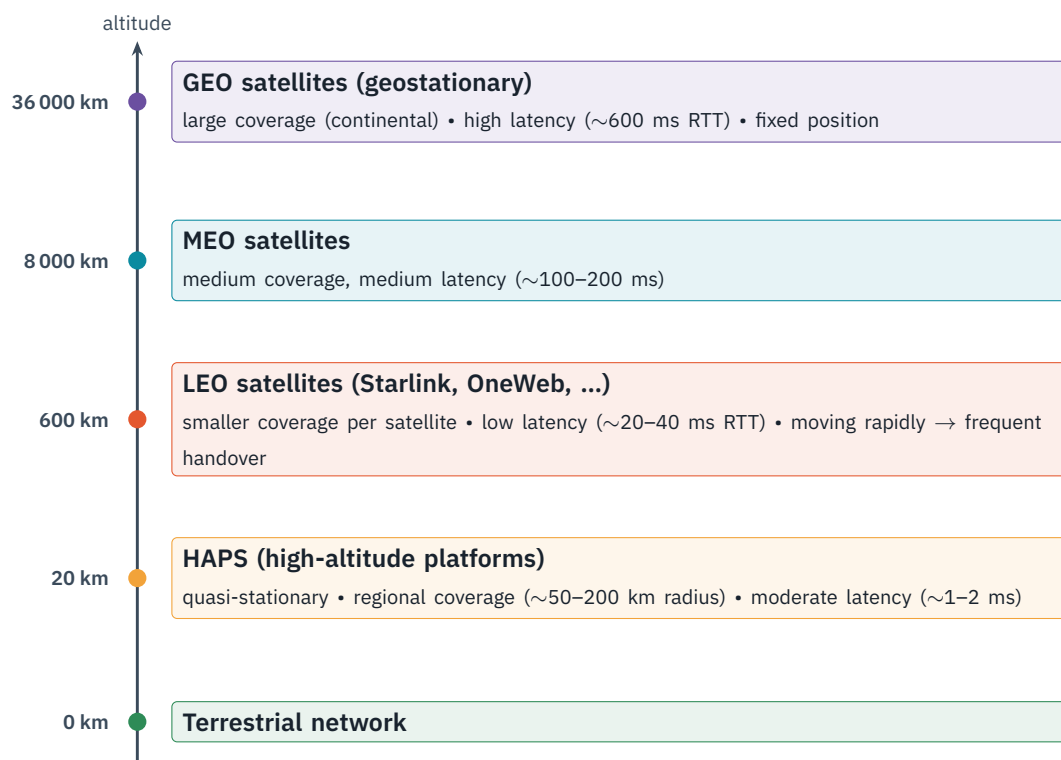


Figure 26.8 Non-terrestrial network platforms by altitude

3GPP Standardization Progress

Release	Feature	Status
R15-R16	Study items (TR 38.811, 38.821)	Completed
R17	Basic NR-NTN (transparent payload, GEO/LEO)	STANDARDIZED
R17	IoT-NTN (NB-IoT/eMTC over satellite)	STANDARDIZED
R18	NTN enhancements (mobility, coverage)	STANDARDIZED
R18	NR-NTN on FR2 (above 10 GHz)	Study
R19	Regenerative payloads, store-and-forward	Study
R19	Direct-to-cell (smartphone satellite)	Study/WI

Key Technical Challenges Solved in R17

1. **Timing advance:** Large propagation delay (up to 540 ms for GEO) → pre-compensation using GNSS at UE
2. **Doppler shift:** LEO at 7.5 km/s → up to ± 24 ppm at S-band → pre-compensation
3. **Large cells:** Satellite beams cover 100s of km → timing advance differential within cell
4. **HARQ:** Disabled or modified due to long RTT (feedback arrives too late)

Architecture Options

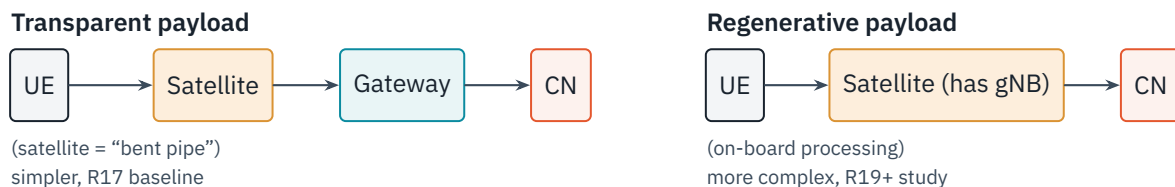


Figure 26.9 Transparent versus regenerative satellite payload

Benefits

- Ubiquitous global coverage
- Disaster resilience and emergency communication
- Direct-to-cell: unmodified smartphones connect to satellites
- IoT coverage in remote areas (agriculture, shipping, pipelines)
- Network redundancy and multi-connectivity

Challenges

- **Link budget:** Very long distances require large antenna gains
- **Spectrum coordination:** Coexistence with terrestrial networks
- **Handover:** LEO satellites move rapidly (4-8 min visibility window)
- **Interference:** Satellite beams overlap with terrestrial cells

- **Capacity:** Limited per-user throughput from satellite
- **Constellation cost:** Thousands of satellites needed for global LEO coverage

Relation to 5G

- NTN is PART of 5G-Advanced (R17-R18 specifications exist)
- Direct-to-cell (smartphone-satellite) commercially launching (T-Mobile/SpaceX, AST SpaceMobile)
- 6G targets tighter integration: seamless terrestrial/NTN handover, multi-layer NTN

Open Questions

- How to achieve seamless handover between LEO satellites?
- Regenerative vs transparent payload — which wins economically?
- How much capacity can satellite provide per user?
- Integration with terrestrial for load balancing in dense areas?

8 Semantic Communications

Status: ● RESEARCH / ● SPECULATIVE

What

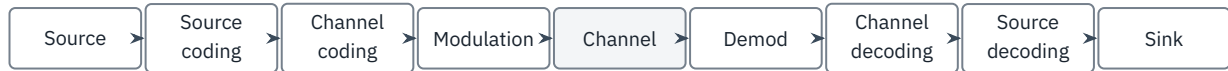
Semantic Communications transmit the **meaning** (semantics) of information rather than exact bit sequences. Instead of ensuring every bit is correctly received, the system ensures the intended meaning or task outcome is preserved, potentially requiring far fewer transmitted bits.

Why / Problem Addressed

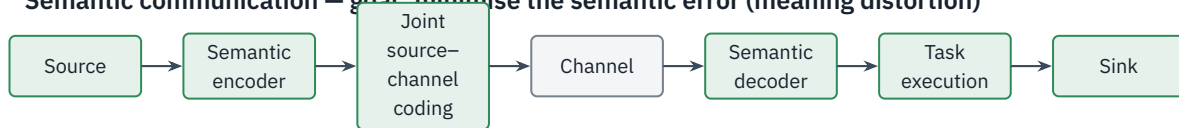
- Shannon's theory optimizes bit transmission, not information utility
- Most transmitted data has redundancy that classical coding doesn't exploit
- Video call: 90%+ of pixels don't change between frames
- Machine-to-machine communication doesn't need human-readable formats
- Approaching Shannon capacity limits → need new paradigm for further gains

Conceptual Framework

Classical communication – goal: minimise the BER



Semantic communication – goal: minimise the semantic error (meaning distortion)



Example (image transmission): classical – transmit all pixels, recover all pixels; semantic – extract key features, transmit a compact representation, reconstruct the image or classify directly.

Figure 26.10 Classical versus semantic communication

Key Technologies

1. Deep Joint Source-Channel Coding (DeepJSCC)

- Single neural network replaces separate source and channel coding
- Graceful degradation (no cliff effect)
- Adapts to channel quality continuously

2. Knowledge Graphs / Shared Knowledge Base

- Transmitter and receiver share background knowledge
- Only novel/unexpected information needs transmission
- Dramatic compression for known contexts

3. Task-Oriented Communication

- Optimize for downstream task (classification, control) not reconstruction
- Transmit only task-relevant features
- Example: Autonomous driving → transmit object positions, not full video

Potential Bandwidth Savings

Application	Classical	Semantic	Reduction
Text (translation task)	Full sentence bits	Intent + entities	10-100×
Image (classification)	All pixels (JPEG)	Feature vector	100-1000×
Video (surveillance)	Full frames (H.265)	Events + changes	10-50×
Control (robotics)	Sensor data stream	Action commands	50-500×

Note: These are theoretical/experimental estimates, not proven at scale

Benefits

- Extreme compression beyond Shannon limits for specific tasks
- Graceful degradation under poor channel conditions
- Natural integration with AI at endpoints
- Efficient machine-to-machine communication
- Reduced latency (fewer bits to transmit)

Challenges

- **No mathematical framework:** Shannon theory doesn't cover semantics
- **Shared knowledge assumption:** Tx and Rx must agree on semantic model
- **Generalization:** Different tasks need different semantic encoders
- **Security:** Semantic attacks (adversarial examples that preserve bits but corrupt meaning)
- **Standardization:** How to specify semantic protocols?
- **Backward compatibility:** Cannot coexist with classical systems easily
- **Evaluation metrics:** What replaces BER/BLER for semantic systems?

Relation to 5G

- 5G has no semantic communication features
- Conceptually orthogonal to current standards
- May first appear in application layer (not radio standard)
- Could eventually change fundamental PHY design in 6G+

Open Questions

- Is a universal semantic language possible, or always task-specific?
- How to handle unknown/novel semantic content?
- Can semantic and classical communication coexist on same channel?
- Who defines the semantic model — standards body, AI training?
- Timeline: Is this 6G or 7G technology?

9 Digital Twins for Networks

■ Status: ● RESEARCH

■ What

A Network Digital Twin is a real-time, high-fidelity virtual replica of a physical network that continuously synchronizes with the live network. It enables what-if analysis, predictive optimization, and autonomous network management by simulating changes before deploying them.

■ Why / Problem Addressed

- Network optimization is reactive (fix after failure)
- Testing changes on live networks is risky
- 5G/6G network complexity exceeds human ability to manage
- Need for predictive maintenance and proactive optimization
- Training AI/ML models requires safe environments

■ Architecture

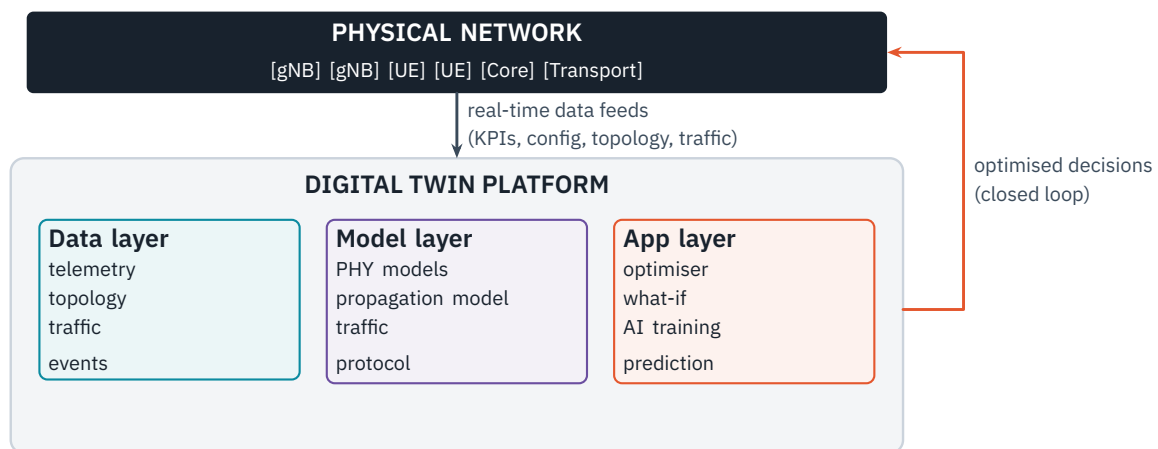


Figure 26.11 Network digital twin in a closed loop

■ Use Cases

1. **Predictive maintenance:** Detect degradation before failure
2. **Configuration optimization:** Test parameter changes in simulation
3. **Capacity planning:** Simulate future growth scenarios
4. **AI/ML training:** Generate synthetic training data safely
5. **Anomaly detection:** Compare twin prediction vs reality
6. **Disaster recovery:** Pre-plan failover configurations
7. **Network slicing:** Simulate SLA compliance before provisioning

I Benefits

- Risk-free experimentation on virtual network
- Continuous optimization without service disruption
- Faster AI model development and validation
- Reduced OPEX through automation and prediction
- Holistic network visibility

I Challenges

- **Fidelity:** Twin must accurately represent physical network behavior
- **Real-time synchronization:** Data collection and model update latency
- **Scale:** Modeling millions of devices in real-time is computationally expensive
- **Validation:** How to verify the twin matches reality?
- **Data requirements:** Enormous telemetry data streams
- **Standardization:** No agreed data model or interface standard yet

I Relation to 5G

- 5G O-RAN provides data interfaces (E2, O1, O2) that feed twins
- 5G network automation (SON) is a simpler precursor
- Some vendors offer “digital twin” products today (mostly offline simulation)
- 6G targets real-time, closed-loop digital twins

I Open Questions

- What level of fidelity is “good enough” for different use cases?
- How to handle model uncertainty and validation?
- Centralized vs distributed twin architecture?
- Standardized data models and APIs?

10 Advanced Positioning (cm-Level)

■ Status: ● STUDY ITEM (Release 18 positioning enhancements)

■ What

6G targets centimeter-level (1-10 cm) positioning accuracy using joint communication and positioning techniques, advanced signal processing, and AI/ML assistance. This goes far beyond GPS (~1-5 m) and 5G R16 positioning (~1 m outdoor).

■ Why / Problem Addressed

- Indoor positioning remains unsolved (GPS unavailable)
- Industrial automation requires mm-cm precision
- Autonomous systems (drones, robots, vehicles) need precise localization
- AR/XR applications require 6DoF tracking at cm level
- Asset tracking in warehouses, hospitals, factories

■ Positioning Techniques Evolution

Release	Accuracy	Techniques
R16 (5G baseline)	~3–10 m	DL-TDoA, UL-TDoA, multi-RTT
R17 (enhancement)	~0.5–3 m	Improved DL-PRS, NR positioning
R18 (study)	~0.2–1 m	Carrier phase, sidelink, AI/ML
R19+ / 6G target	~0.01–0.1 m	cm-level, 6DoF

■ Key Technologies for cm-Level Accuracy

■ Carrier-Phase Positioning

- **Standard ranging** uses the signal envelope: resolution $\sim c/BW$; for $BW = 100$ MHz the resolution is about 3 m (before processing gain).
- **Carrier-phase ranging** uses the carrier wavelength λ : at 3.5 GHz, $\lambda = 8.6$ cm (sub-cm potential); at 28 GHz, $\lambda = 1.07$ cm (mm potential).
- **Challenge:** integer-ambiguity resolution. The measured phase is

$$\varphi = \frac{2\pi}{\lambda} d + 2\pi N + \text{noise}$$

and N must be resolved to obtain the absolute distance.

■ Multi-RTT with Large Bandwidth

- Round-Trip-Time measurement with 400 MHz - 2 GHz bandwidth

- Sub-nanosecond timing resolution
- Multiple base stations for triangulation

■ AI/ML-Assisted Positioning

- **Fingerprinting:** ML models map channel features to position
- **NLOS identification:** AI classifies LoS vs NLoS paths
- **Sensor fusion:** Combine radio with IMU, camera, LiDAR
- **Channel charting:** Unsupervised learning of radio geometry

■ Joint Communication and Positioning

- Same waveform serves dual purpose
- PRS (Positioning Reference Signals) integrated with data transmission
- ISAC synergy: sensing capabilities enhance positioning

■ Accuracy vs Technique Comparison

Technique	Accuracy	Pros	Cons
DL-TDoA (R16)	3-10 m	Standard, UE-based	Clock sync needed
Multi-RTT (R16)	1-5 m	No sync needed	Multiple gNBs
DL-AoD (R16)	5-15 m	Single gNB possible	Antenna cal needed
Enhanced (R17)	0.5-3 m	Improved PRS	Bandwidth limited
Carrier-phase (R18)	1-10 cm	Very high accuracy	Ambiguity resolution
AI-assisted (R18+)	0.3-3 m	Works in NLOS	Training data needed
6G target	1-10 cm	Native, real-time	Full system redesign

■ 3GPP Progress

- **R16:** Basic NR positioning (TS 38.305, 38.455)
- **R17:** Positioning enhancements (improved accuracy, reduced latency)
- **R18:** Study on AI/ML for positioning, carrier-phase, sidelink positioning
- **R18:** Integrity and reliability requirements for positioning
- **R19:** Further accuracy improvements, ISAC-based positioning

I Benefits

- Ubiquitous cm-level positioning (indoor and outdoor)
- No additional infrastructure (reuses communication network)
- 6DoF tracking (position + orientation) for XR/AR
- Enables new applications: digital twins, autonomous navigation

I Challenges

- **Carrier-phase ambiguity:** Integer ambiguity resolution is hard in mobile scenarios
- **Multipath:** Reflections corrupt ranging measurements
- **NLOS:** Dominant in indoor environments
- **Synchronization:** Network timing accuracy limits achievable positioning
- **UE complexity:** Processing requirements for high-accuracy positioning
- **Power consumption:** Continuous positioning drains mobile battery

I Relation to 5G

- 5G NR positioning (R16-R17) provides the foundation
- Each release incrementally improves accuracy
- 6G integrates positioning as a native service (not an add-on)
- Carrier-phase and AI techniques studied in R18 for potential R19 specification

I Open Questions

- Can carrier-phase work reliably in mobile environments?
- How to handle integer ambiguity in real-time with mobility?
- What infrastructure density is needed for cm-level indoor coverage?
- How to balance positioning accuracy vs communication performance?

Technology Maturity Matrix

#	Technology	TRL*	3GPP Status	Standardization Timeline
1	AI-Native Networks / AI-RAN	4-5	Study Item (R18 TR 38.843)	R19-R20 Work Items
2	XL-MIMO	2-3	Not yet studied	R21+ (6G)
3	Cell-Free Massive MIMO	3-4	Not yet studied (CoMP precursor)	R20-R21
4	ISAC	3-4	Study Item (R19)	R20 Work Items
5	RIS	3-4	Study Item (R18-19 TR 38.857)	R20 Work Items
6	Sub-THz Communications	2-3	Not yet studied	R21+ (6G)
7	NTN (basic)	7-8	Standardized (R17)	Deployed
7	NTN (advanced)	4-5	Study/WI (R18-19)	R19-R20
8	Semantic Communications	1-2	Not studied	R22+ (post-6G?)
9	Digital Twins for Networks	3-4	Not yet studied	R20-R21
10	Advanced Positioning (cm)	4-5	Study Item (R18)	R19-R20

*TRL = Technology Readiness Level (1 = basic research, 9 = deployed)

TRL Scale Reference

TRL	Meaning
1	Basic principles observed
2	Technology concept formulated
3	Experimental proof of concept
4	Technology validated in the lab
5	Technology validated in a relevant environment
6	Technology demonstrated in a relevant environment
7	System prototype demonstrated in an operational environment
8	System complete and qualified
9	System proven in an operational environment (deployed)

Projected Timeline: 3GPP Releases and 6G Technologies

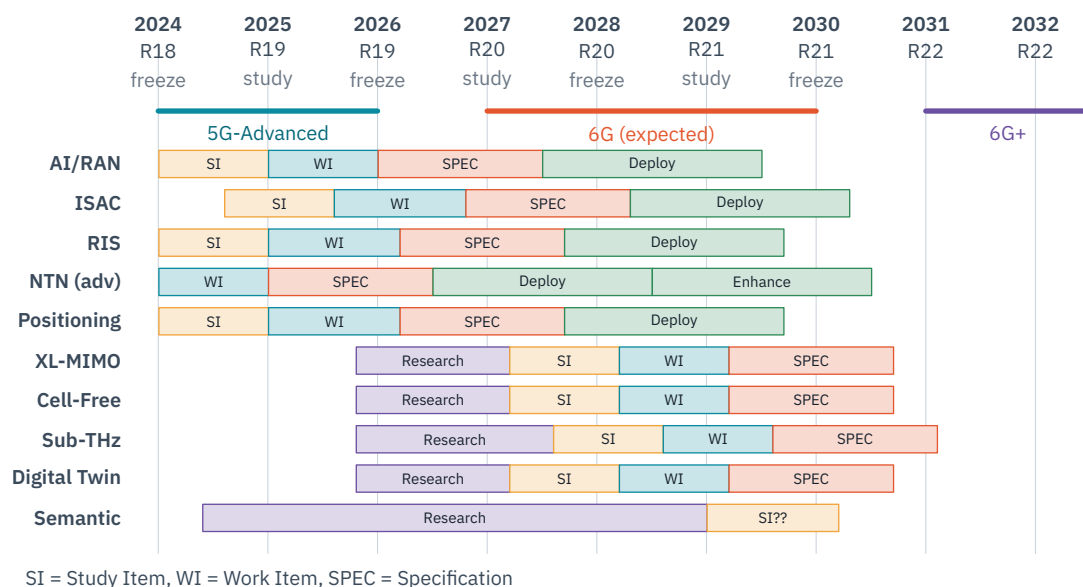


Figure 26.12 Projected timeline of 3GPP releases and 6G technologies

What to Expect per Release

Release	Timeline	Key 6G-Related Features
R19 (5G-Adv)	2025-2026	ISAC study, advanced NTN, positioning enhancements, AI/RAN WI
R20 (early 6G?)	2027-2028	ISAC specification, RIS specification, cell-free studies, sub-THz studies
R21 (6G)	2029-2030	XL-MIMO, sub-THz, semantic study, full AI-native, digital twin
R22 (6G+)	2031-2032	Semantic communication, advanced THz, autonomous networks

Summary: 5G vs 6G Paradigm Shifts

Aspect	5G	6G
AI role	Optimization add-on	Native design principle
Spectrum	Sub-6 + mmWave (FR1+FR2)	+ Sub-THz (FR3?)
Architecture	Cell-based	Cell-free + NTN
Coverage	Terrestrial	Terrestrial + space + aerial
Sensing	Not supported	Integrated (ISAC)
Positioning	~1m accuracy	~1 cm accuracy
Environment control	Passive (adapt to channel)	Active (RIS shapes channel)
Communication goal	Transmit bits reliably	Transmit meaning efficiently
Network management	Human-configured	Autonomous (digital twin + AI)
Antenna scale	64-256 elements	1000-10000+ elements
Peak rate	20 Gbps	200+ Gbps
Latency	1 ms	0.1 ms

Key Takeaways

1. **Not all “6G technologies” are equal** — NTN is already standardized, while semantic communication remains speculative. Always check maturity level.
2. **6G is evolutionary AND revolutionary** — Some features (AI-RAN, ISAC, RIS) evolve from 5G-Advanced. Others (semantic, THz, cell-free) represent paradigm shifts.
3. **3GPP drives standardization** — Technologies must pass: Research → Study Item → Work Item → Technical Specification. This process takes 3-5 years per technology.
4. **Timing: 6G specification expected ~2028-2030**, with first deployments ~2030-2032.
5. **IMT-2030 framework** sets targets but actual specifications may differ from initial visions (as happened with IMT-2020/5G).
6. **Economic viability** ultimately determines adoption — technically superior solutions may lose to “good enough” practical alternatives.

References and Further Reading

■ 3GPP Documents

- TR 38.843: Study on AI/ML for NR air interface (R18)
- TR 38.857: Study on NR network-controlled repeaters (R18)
- TR 38.811: Study on NR to support NTN (R15)
- TR 38.821: Solutions for NR to support NTN (R16)
- TS 38.305: Stage 2 functional specification for positioning (R16+)

■ ITU-R

- IMT-2030 Framework Recommendation (ITU-R M.2160)
- Report ITU-R M.2516: Future technology trends for IMT systems toward 2030+

■ Key Research Papers

- Björnson et al., “Scalable Cell-Free Massive MIMO Systems,” IEEE Trans. Commun., 2020
- Di Renzo et al., “Smart Radio Environments Empowered by RIS,” JSAC, 2020
- Rappaport et al., “Wireless Communications and Applications Above 100 GHz,” IEEE Access, 2019
- Qin et al., “Semantic Communications: Principles and Challenges,” arXiv, 2021
- Liu et al., “Integrated Sensing and Communications,” IEEE JSAC, 2022
- Amiri et al., “Extremely Large MIMO,” IEEE Commun. Mag., 2024

■ Industry Roadmaps

- Next G Alliance (ATIS): 6G Roadmap
- Hexa-X / Hexa-X-II (EU 6G Flagship Project)
- Samsung 6G Vision (2020, updated 2023)
- Nokia Bell Labs 6G Vision
- NGMN Alliance: 6G Drivers and Vision

Module 26 — Last updated: August 2026

Note: This is a living document. Technology maturity assessments will change as research progresses and 3GPP work advances.